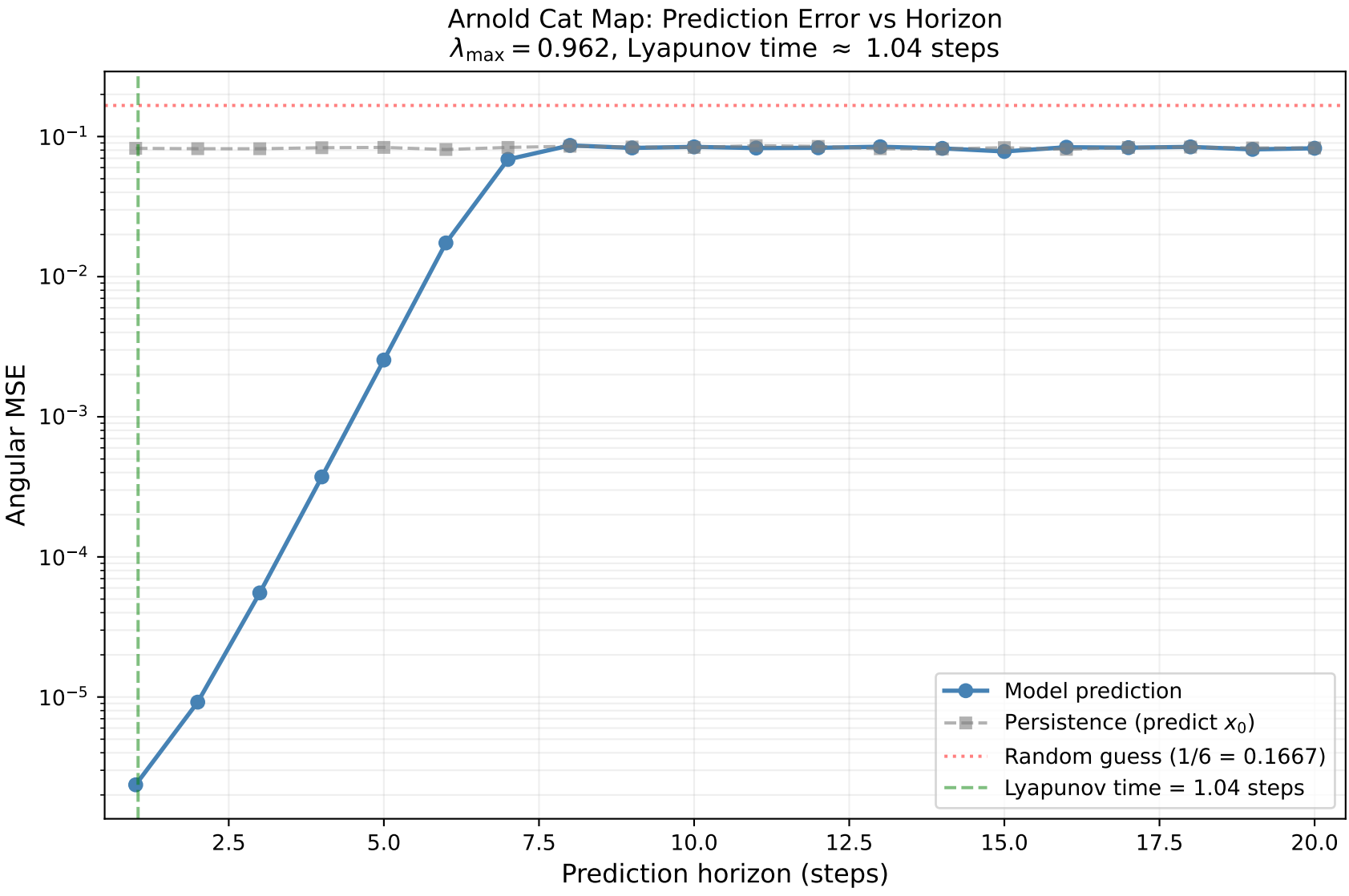
Arnold Cat Map: Prediction Error vs Horizon
$\lambda_{max} = 0.962$, Lyapunov time $\approx$ 1.04 steps
Angular MSE
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Prediction horizon (steps)
Model prediction
Persistence (predict x_0)
Random guess (1/6 = 0.1667)
Lyapunov time = 1.04 steps